

LINEAR ALGEBRA DONE RIGHT, 4TH ED. BY SHELDON AXLER,
PROBLEM 1 FROM CHAPTER 3C

Problem. Suppose $T \in \mathcal{L}(V, W)$. Show that with respect to each choice of bases of V and W , the matrix of T has at least $\dim \text{range } T$ nonzero entries.

Solution. Let $v_1, \dots, v_n$ be a basis of V and $w_1, \dots, w_m$ be a basis of W , and let M denote the matrix of T with respect to these bases. By definition, the j -th column of M consists of the scalars $A_{1,j}, \dots, A_{m,j}$ such that

$$T(v_j) = A_{1,j} w_1 + \cdots + A_{m,j} w_m.$$

The span of the columns of M therefore corresponds exactly to the set of all linear combinations of $T(v_1), \dots, T(v_n)$, which is $\text{range } T$. Consequently, the number of linearly independent columns of M equals

$$\dim \text{range } T =: k.$$

Hence M has at least k linearly independent columns. Each such column must be nonzero, since the zero vector cannot belong to any linearly independent list. A nonzero column contains at least one nonzero entry, and entries belong to exactly one column, so these k columns contribute at least k *distinct* nonzero entries to M .

Therefore the matrix of T has at least $\dim \text{range } T$ nonzero entries. ■